

S1 — Hyperparameter Search Grids

This supplementary file is **auto-generated** by `dropout_bench_v3_G_13_supplementary_hyperparameter_grids.py`. It documents the complete candidate search grids for all tuned models in the benchmark. For Family B, grids are read from JSON metadata files produced at pipeline run-time. For Family A, grids are extracted from the pipeline source constants via AST parsing.

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Family B — Static Early-Window (Comparable Arm)

All Family B models use enrollment-level early-window features. Validation uses a 20% enrollment-level hold-out (`GroupShuffleSplit`) stratified by event, except Neural-MTLR and DeepHit which use a 10% internal fraction.

Cox Comparable

Candidates: 24 | **Selection criterion:** highest validation C-index

Candidate	penalizer	l1_ratio
1	0.001	0.0
2	0.001	0.25
3	0.001	0.5
4	0.001	0.75
5	0.01	0.0
6	0.01	0.25
7	0.01	0.5
8	0.01	0.75
9	0.05	0.0
10	0.05	0.25
11	0.05	0.5
12	0.05	0.75
13	0.1	0.0
14	0.1	0.25
15	0.1	0.5
16	0.1	0.75
17	0.2	0.0
18	0.2	0.25
19	0.2	0.5
20	0.2	0.75

Candidate	penalizer	l1_ratio
21	0.5	0.0
22	0.5	0.25
23	0.5	0.5
24	0.5	0.75

DeepSurv

Candidates: 24 | **Selection criterion:** lowest validation loss (early stopping, patience=10) *Architecture grid: [32,16], [64,32], [128,64]. Early stopping on val loss.*

Candidate	hidden_dims	dropout	learning_rate	weight_decay
1	[32, 16]	0.1	5e-04	1e-05
2	[32, 16]	0.1	5e-04	1e-04
3	[32, 16]	0.1	0.001	1e-05
4	[32, 16]	0.1	0.001	1e-04
5	[32, 16]	0.3	5e-04	1e-05
6	[32, 16]	0.3	5e-04	1e-04
7	[32, 16]	0.3	0.001	1e-05
8	[32, 16]	0.3	0.001	1e-04
9	[64, 32]	0.1	5e-04	1e-05
10	[64, 32]	0.1	5e-04	1e-04
11	[64, 32]	0.1	0.001	1e-05
12	[64, 32]	0.1	0.001	1e-04
13	[64, 32]	0.3	5e-04	1e-05
14	[64, 32]	0.3	5e-04	1e-04
15	[64, 32]	0.3	0.001	1e-05
16	[64, 32]	0.3	0.001	1e-04
17	[128, 64]	0.1	5e-04	1e-05
18	[128, 64]	0.1	5e-04	1e-04
19	[128, 64]	0.1	0.001	1e-05
20	[128, 64]	0.1	0.001	1e-04
21	[128, 64]	0.3	5e-04	1e-05
22	[128, 64]	0.3	5e-04	1e-04
23	[128, 64]	0.3	0.001	1e-05
24	[128, 64]	0.3	0.001	1e-04

Random Survival Forest

Candidates: 24 | **Selection criterion:** lowest validation IBS

Candidate	n_estimators	min_samples_leaf	max_depth	max_features
1	200	3	None	sqrt
2	300	5	None	sqrt
3	400	3	None	sqrt
4	200	10	None	sqrt
5	300	20	None	sqrt
6	400	5	None	0.8
7	200	3	None	0.7
8	300	10	None	0.8
9	200	3	8	sqrt
10	300	5	8	sqrt
11	400	3	8	sqrt
12	200	10	8	0.8
13	300	20	8	0.8
14	400	5	8	0.7
15	200	3	8	0.7
16	300	10	8	0.7
17	200	3	15	sqrt
18	300	5	15	sqrt
19	400	3	15	sqrt
20	200	10	15	0.8
21	300	20	15	0.8
22	400	5	15	0.7
23	200	3	15	0.7
24	300	10	15	0.7

Neural-MTLR

Candidates: 24 | **Selection criterion:** lowest validation IBS (early stopping, patience=8) *num_durations* controls survival-time discretization. *Early stopping on val IBS.*

Candidate	num_durations	hidden_dims	dropout	learning_rate	weight_decay	batch_size
1	20	[64, 32]	0.1	0.005	1e-05	512
2	20	[64, 32]	0.1	0.01	1e-05	512

Candidate	num_durations	hidden_dims	dropout	learning_rate	weight_decay	batch_size
3	20	[64, 32]	0.2	0.005	1e-04	512
4	20	[64, 32]	0.2	0.01	1e-04	512
5	20	[128, 64]	0.1	0.005	1e-05	512
6	20	[128, 64]	0.1	0.01	1e-05	512
7	20	[128, 64]	0.2	0.005	1e-04	512
8	20	[128, 64]	0.2	0.01	1e-04	512
9	20	[256, 128]	0.1	0.005	1e-05	512
10	20	[256, 128]	0.2	0.005	1e-04	512
11	20	[256, 128]	0.2	0.01	1e-04	1024
12	20	[256, 128]	0.3	0.001	1e-04	1024
13	40	[64, 32]	0.1	0.005	1e-05	512
14	40	[64, 32]	0.2	0.005	1e-04	512
15	40	[64, 32]	0.2	0.01	1e-04	512
16	40	[64, 32]	0.3	0.001	1e-04	512
17	40	[128, 64]	0.1	0.005	1e-05	512
18	40	[128, 64]	0.1	0.01	1e-05	512
19	40	[128, 64]	0.2	0.005	1e-04	512
20	40	[128, 64]	0.2	0.001	1e-04	1024
21	40	[256, 128]	0.1	0.005	1e-05	512
22	40	[256, 128]	0.2	0.005	1e-04	512
23	40	[256, 128]	0.2	0.01	1e-04	1024
24	40	[256, 128]	0.3	0.001	1e-04	1024

DeepHit

Candidates: 24 | **Selection criterion:** lowest validation IBS (early stopping, patience=8) *alpha/sigma are DeepHit-specific ranking loss weights.*

Candidate	num_durations	hidden_dims	dropout	alpha	sigma	learning_rate	weight_decay	batch_size
1	20	[64, 32]	0.1	0.2	0.1	0.005	1e-05	512
2	20	[64, 32]	0.1	0.2	0.1	0.01	1e-05	512
3	20	[64, 32]	0.2	0.3	0.1	0.005	1e-04	512
4	20	[64, 32]	0.2	0.3	0.2	0.01	1e-04	512
5	20	[128, 64]	0.1	0.2	0.1	0.005	1e-05	512
6	20	[128, 64]	0.1	0.2	0.2	0.01	1e-05	512

Candidate	num_durations	hidden_dims	dropout	alpha	sigma	learning_rate	weight_decay	batch_size
7	20	[128, 64]	0.2	0.3	0.1	0.005	1e-04	512
8	20	[128, 64]	0.2	0.3	0.2	0.001	1e-04	1024
9	20	[256, 128]	0.1	0.2	0.1	0.005	1e-05	512
10	20	[256, 128]	0.2	0.3	0.2	0.001	1e-04	1024
11	20	[256, 128]	0.2	0.3	0.1	0.005	1e-04	512
12	20	[256, 128]	0.3	0.2	0.1	0.001	1e-04	1024
13	40	[64, 32]	0.1	0.2	0.1	0.005	1e-05	512
14	40	[64, 32]	0.1	0.2	0.2	0.01	1e-05	512
15	40	[64, 32]	0.2	0.3	0.1	0.005	1e-04	512
16	40	[64, 32]	0.2	0.3	0.2	0.01	1e-04	512
17	40	[128, 64]	0.1	0.2	0.1	0.005	1e-05	512
18	40	[128, 64]	0.1	0.2	0.2	0.01	1e-05	512
19	40	[128, 64]	0.2	0.3	0.1	0.005	1e-04	512
20	40	[128, 64]	0.2	0.3	0.2	0.001	1e-04	1024
21	40	[256, 128]	0.1	0.2	0.1	0.005	1e-05	512
22	40	[256, 128]	0.2	0.3	0.1	0.005	1e-04	512
23	40	[256, 128]	0.2	0.3	0.2	0.01	1e-04	1024
24	40	[256, 128]	0.3	0.2	0.1	0.001	1e-04	1024

GB Cox

Candidates: 24 | **Selection criterion:** lowest validation IBS

Candidate	loss	learning_rate	n_estimators	max_depth	subsample
1	coxph	0.02	100	1	1.0
2	coxph	0.03	150	1	1.0
3	coxph	0.05	150	1	1.0
4	coxph	0.05	250	1	0.8
5	coxph	0.08	150	1	0.8
6	coxph	0.1	100	1	1.0

Candidate	loss	learning_rate	n_estimators	max_depth	subsample
7	coxph	0.1	150	1	0.8
8	coxph	0.1	300	1	0.8
9	coxph	0.02	200	2	1.0
10	coxph	0.03	200	2	1.0
11	coxph	0.05	200	2	1.0
12	coxph	0.05	250	2	0.8
13	coxph	0.08	200	2	0.8
14	coxph	0.08	250	2	0.8
15	coxph	0.1	150	2	1.0
16	coxph	0.1	300	2	0.8
17	coxph	0.02	300	3	1.0
18	coxph	0.03	250	3	1.0
19	coxph	0.05	200	3	0.8
20	coxph	0.05	300	3	1.0
21	coxph	0.08	150	3	0.8
22	coxph	0.08	300	3	0.8
23	coxph	0.1	100	3	0.8
24	coxph	0.1	200	3	0.8

Royston-Parmar

Candidates: 24 | Selection criterion: lowest validation IBS

Candidate	n_baseline_knots	penalizer	l1_ratio
1	2	0.01	0.0
2	2	0.025	0.0
3	2	0.05	0.0
4	2	0.1	0.0
5	2	0.15	0.0
6	2	0.2	0.0
7	2	0.3	0.0
8	2	0.5	0.0
9	2	0.7	0.0
10	2	1.0	0.0
11	2	2.0	0.0
12	2	5.0	0.0

Candidate	n_baseline_knots	penalizer	l1_ratio
13	3	0.01	0.0
14	3	0.025	0.0
15	3	0.05	0.0
16	3	0.1	0.0
17	3	0.15	0.0
18	3	0.2	0.0
19	3	0.3	0.0
20	3	0.5	0.0
21	3	0.7	0.0
22	3	1.0	0.0
23	3	2.0	0.0
24	3	5.0	0.0

Weibull AFT

Candidates: 24 | Selection criterion: lowest validation IBS

Candidate	penalizer	l1_ratio
1	1e-04	0.0
2	2e-04	0.0
3	5e-04	0.0
4	0.001	0.0
5	0.002	0.0
6	0.005	0.0
7	0.01	0.0
8	0.02	0.0
9	0.05	0.0
10	0.1	0.0
11	0.15	0.0
12	0.2	0.0
13	0.3	0.0
14	0.4	0.0
15	0.5	0.0
16	0.6	0.0
17	0.7	0.0
18	0.8	0.0

Candidate	penalizer	l1_ratio
19	1.0	0.0
20	1.5	0.0
21	2.0	0.0
22	3.0	0.0
23	4.0	0.0
24	5.0	0.0

XGBoost AFT

Candidates: 24 | Selection criterion: lowest validation IBS

Candidate	aft_loss_distribution	aft_loss_distribution_scale	learning_rate	num_boost_round	max_depth	min_child_weight	subsample	colsample_bytree	reg_lambda	reg_alpha
1	logistic	1.0	0.03	200	2	1.0	0.8	0.8	1.0	0.0
2	logistic	1.0	0.05	200	2	1.0	0.8	0.8	1.0	0.0
3	logistic	1.0	0.03	400	2	1.0	0.8	0.8	1.0	0.0
4	logistic	1.0	0.05	400	2	1.0	0.8	0.8	1.0	0.0
5	logistic	1.5	0.03	200	2	1.0	0.8	0.8	1.0	0.0
6	logistic	1.5	0.05	200	2	1.0	0.8	0.8	1.0	0.0
7	logistic	1.5	0.03	400	2	2.0	0.8	0.8	5.0	0.0
8	logistic	1.5	0.05	400	2	2.0	0.8	0.8	5.0	0.0
9	logistic	2.0	0.03	200	2	1.0	0.8	0.8	1.0	0.0
10	logistic	2.0	0.05	200	2	1.0	0.8	0.8	1.0	0.0
11	logistic	2.0	0.03	400	2	2.0	0.8	0.8	5.0	0.0
12	logistic	2.0	0.05	400	2	2.0	0.8	0.8	5.0	0.0
13	normal	1.0	0.03	200	2	1.0	0.8	0.8	1.0	0.0
14	normal	1.0	0.05	200	2	1.0	0.8	0.8	1.0	0.0
15	normal	1.0	0.03	400	2	1.0	0.8	0.8	1.0	0.0
16	normal	1.0	0.05	400	2	1.0	0.8	0.8	1.0	0.0
17	normal	1.5	0.03	200	2	1.0	0.8	0.8	1.0	0.0
18	normal	1.5	0.05	200	2	1.0	0.8	0.8	1.0	0.0
19	normal	1.5	0.03	400	2	2.0	0.8	0.8	5.0	0.0
20	normal	1.5	0.05	400	2	2.0	0.8	0.8	5.0	0.0
21	normal	2.0	0.03	200	2	1.0	0.8	0.8	1.0	0.0
22	normal	2.0	0.05	200	2	1.0	0.8	0.8	1.0	0.0

Candidate	aft_loss_distribution	aft_loss_distribution_scale	learning_rate	num_boost_round	max_depth	min_child_weight	subsample	colsample_bytree	reg_lambda	reg_alpha
23	normal	2.0	0.03	400	2	2.0	0.8	0.8	5.0	0.0
24	normal	2.0	0.05	400	2	2.0	0.8	0.8	5.0	0.0

Family A — Dynamic Weekly (Person-Period Arm)

All Family A models use person-period rows (one row per enrollment per week). Validation uses a 10–20% enrollment-level GroupShuffleSplit.

Linear Discrete-Time Hazard

Candidates: 24 | **Selection criterion:** highest row-level validation C-index *Penalty type (L1/L2) × C regularization strength.*

Candidate	penalty	C
1	l1	0.001
2	l1	0.005
3	l1	0.01
4	l1	0.05
5	l1	0.1
6	l1	0.5
7	l1	1.0
8	l1	5.0
9	l1	10.0
10	l1	50.0
11	l1	100.0
12	l1	500.0
13	l2	0.001
14	l2	0.005
15	l2	0.01
16	l2	0.05
17	l2	0.1
18	l2	0.5
19	l2	1.0
20	l2	5.0
21	l2	10.0

Candidate	penalty	C
22	l2	50.0
23	l2	100.0
24	l2	500.0

Poisson Piecewise-Exponential

Candidates: 24 | **Selection criterion:** lowest validation log-loss on discrete hazard 1-D grid over $L2$ regularization strength (α). Single-param sweep via statsmodels GLM.

Candidate	alpha
1	1e-05
2	2e-05
3	5e-05
4	1e-04
5	2e-04
6	5e-04
7	0.001
8	0.002
9	0.005
10	0.01
11	0.02
12	0.05
13	0.1
14	0.2
15	0.3
16	0.5
17	0.7
18	1.0
19	2.0
20	5.0
21	10.0
22	20.0
23	50.0
24	100.0

GB Weekly Hazard

Candidates: 24 | **Selection criterion:** lowest validation log-loss (early stopping) *HistGradientBoostingClassifier with early stopping. Row budget capped by feature count.*

Candidate	max_depth	learning_rate	max_iter	min_samples_leaf	l2_regularization
1	2	0.02	80	300	5.0
2	2	0.02	150	200	5.0
3	2	0.03	120	300	5.0
4	2	0.05	120	300	1.0
5	2	0.05	200	160	5.0
6	2	0.08	150	200	5.0
7	2	0.1	100	250	5.0
8	2	0.1	200	160	10.0
9	3	0.02	150	300	10.0
10	3	0.03	120	200	5.0
11	3	0.03	200	160	5.0
12	3	0.05	120	160	5.0
13	3	0.05	200	200	10.0
14	3	0.08	100	300	5.0
15	3	0.08	200	160	10.0
16	3	0.1	80	300	10.0
17	4	0.02	200	300	10.0
18	4	0.03	150	250	10.0
19	4	0.05	100	300	10.0
20	4	0.05	200	200	10.0
21	4	0.08	80	300	5.0
22	4	0.1	80	200	10.0
23	4	0.1	150	160	10.0
24	4	0.1	250	160	10.0

Neural Discrete-Time

Candidates: 24 | **Selection criterion:** lowest validation loss (early stopping, patience=5) *Grid product: 3 arch $\times$ 2 dropout $\times$ 2 lr $\times$ 2 wd = 24 candidates. Override: candidate_id=9 uses dropout=0.05.*

Candidate	hidden_dims	dropout	learning_rate	weight_decay
1	[32, 16]	0.1	0.001	1e-05
2	[32, 16]	0.1	0.001	1e-04
3	[32, 16]	0.1	5e-04	1e-05
4	[32, 16]	0.1	5e-04	1e-04
5	[32, 16]	0.3	0.001	1e-05
6	[32, 16]	0.3	0.001	1e-04
7	[32, 16]	0.3	5e-04	1e-05
8	[32, 16]	0.3	5e-04	1e-04
9	[64, 32]	0.05	0.001	1e-05
10	[64, 32]	0.1	0.001	1e-04
11	[64, 32]	0.1	5e-04	1e-05
12	[64, 32]	0.1	5e-04	1e-04
13	[64, 32]	0.3	0.001	1e-05
14	[64, 32]	0.3	0.001	1e-04
15	[64, 32]	0.3	5e-04	1e-05
16	[64, 32]	0.3	5e-04	1e-04
17	[128, 64]	0.1	0.001	1e-05
18	[128, 64]	0.1	0.001	1e-04
19	[128, 64]	0.1	5e-04	1e-05
20	[128, 64]	0.1	5e-04	1e-04
21	[128, 64]	0.3	0.001	1e-05
22	[128, 64]	0.3	0.001	1e-04
23	[128, 64]	0.3	5e-04	1e-05
24	[128, 64]	0.3	5e-04	1e-04

Notes on tuning scope

Tuning was deliberately controlled rather than exhaustive: the goal was comparable benchmark representatives, not unbounded per-model optimization. All candidates were evaluated under the same preprocessing pipeline (median imputation, constant-missing categoricals, one-hot encoding, standard scaling fitted on training rows only).

All models were subjected to an equivalent tuning budget of 24 candidate evaluations, ensuring that no model benefited from a disproportionately larger search effort. The budget of 24 candidates was set by the most complex model in the benchmark (DeepSurv), whose architecture and regularisation space required the largest grid; all remaining models were expanded to the same ceiling to ensure symmetric search effort.

Full machine-readable candidate records are available
in `outputs_benchmark_survival/metadata/` as JSON files.